Mechanical Work and Power

Practice: Multiple Choice + Free Response

Use the AP Physics 1 equation sheet conventions. Show clear working for free-response numerical parts. Unless otherwise stated, neglect air resistance and take $g = 9.8\ \text{m s}^{-2}$.

Part A · Multiple Choice

1. A constant force acts 30° above the direction of displacement. Which expression gives the work done by that force over distance d?

A. Fd

B. Fd sin30°

C. Fd cos30°

D. 0

2. A student carries a box horizontally at constant height and constant speed. What is the work done on the box by the student's upward support force?

A. Positive

B. Negative

C. Zero

D. Cannot be determined

3. A block slides right on a rough horizontal surface. The work done by kinetic friction on the block is

A. positive

B. negative

C. zero

D. always equal to the work done by gravity

4. A force does +40 J of work on an object while a second force does −15 J and all other forces do zero work. The change in kinetic energy is

A. −55 J

B. −25 J

C. +25 J

D. +55 J

5. The same constant force F moves an object through the same distance d. In trial 1, θ = 0°. In trial 2, θ = 60°. The ratio W_2/W_1 is

A. 1/4

B. 1/2

C. √3/2

D. 2

6. On a graph of F∥ versus displacement, the work done between two positions is determined by

A. the slope

B. the intercept

C. the signed area under the curve

D. the maximum force only

7. A force-displacement graph lies entirely below the horizontal axis over an interval. The work done by that force over the interval is

A. positive

B. negative

C. zero

D. necessarily constant

8. The net work done on an object is zero. Which statement must be true?

A. Its speed is zero

B. Its kinetic energy is unchanged

C. No forces act

D. Its velocity is unchanged

9. A constant horizontal force acts on a box whose speed doubles. At the later instant, the instantaneous power delivered by that force is

A. one-quarter as large

B. half as large

C. twice as large

D. four times as large

10. A person pushes a box with a constant horizontal force while the box moves at constant speed. The mechanical power delivered by the person is

A. zero

B. constant and nonzero

C. increasing

D. decreasing

11. A motor transfers 2400 J of energy in 6.0 s. Its average power is

A. 40 W

B. 400 W

C. 2400 W

D. 14400 W

12. A force is always perpendicular to an object’s instantaneous velocity. The instantaneous power delivered by that force is

A. positive

B. negative

C. zero

D. proportional to v^2

13. Which statement describes a conservative force?

A. Its work always increases kinetic energy

B. Its work depends on path length

C. Its work between two configurations is path-independent

D. It can never do negative work

14. A block travels between the same two points along two paths of different lengths while kinetic friction has the same magnitude. The magnitude of energy dissipated by friction is

A. the same on both paths

B. larger on the longer path

C. larger on the shorter path

D. zero on both paths

15. On a graph of energy transferred E versus time t, the instantaneous power is represented by

A. the area under the graph

B. the slope of the graph

C. the horizontal intercept

D. E/t^2

16. A force F acts at 120° to the displacement. The work done by the force is

A. positive

B. negative

C. zero

D. equal to Fd

17. A 200 N force acts parallel to motion for 3.0 m, then a 100 N force opposite to motion acts for 2.0 m. The total work done by these two forces is

A. −800 J

B. −400 J

C. +400 J

D. +800 J

18. A constant 150 N force acts in the direction of motion while an object moves at 4.0 m s^{-1}. The instantaneous power delivered is

A. 37.5 W

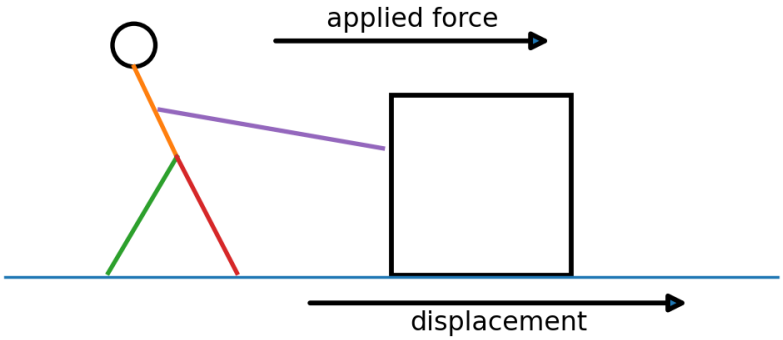
B. 154 W

C. 600 W

D. 2400 W

Part B · Free Response

FRQ 1 · Horizontal push, no friction



A person applies a constant horizontal force of 480 N to a 360 kg box while the box moves forward 3.5 m. Assume there is no friction. The box starts at 1.0 m s^{-1}.

(a) Determine the work done by the person.

(b) Determine the acceleration of the box.

(c) Hence determine its speed after 3.5 m.

(d) Determine the initial kinetic energy.

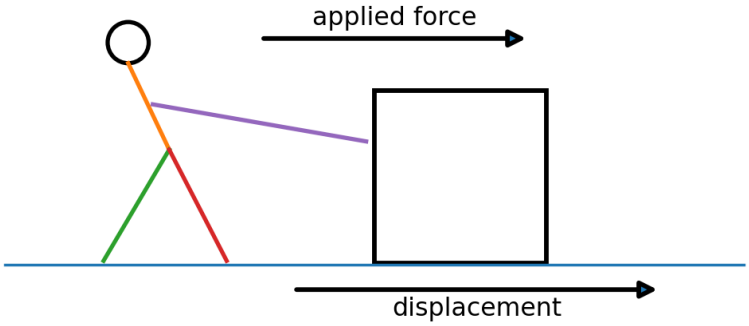
(e) Determine the final kinetic energy.

(f) Determine the gain in kinetic energy.

(g) Determine the average mechanical power delivered by the person over this interval.

(h) Is the person’s instantaneous mechanical power constant? Explain.

FRQ 2 · Horizontal push with friction



A person applies a constant horizontal force of 480 N to a 360 kg box while it moves forward 3.5 m at a constant speed of 1.0 m s^{-1}.

(a) Determine the work done by the person.

(b) Determine the coefficient of kinetic friction and describe the energy transfer taking place.

(c) Determine the average mechanical power delivered by the person. Is the instantaneous power constant?

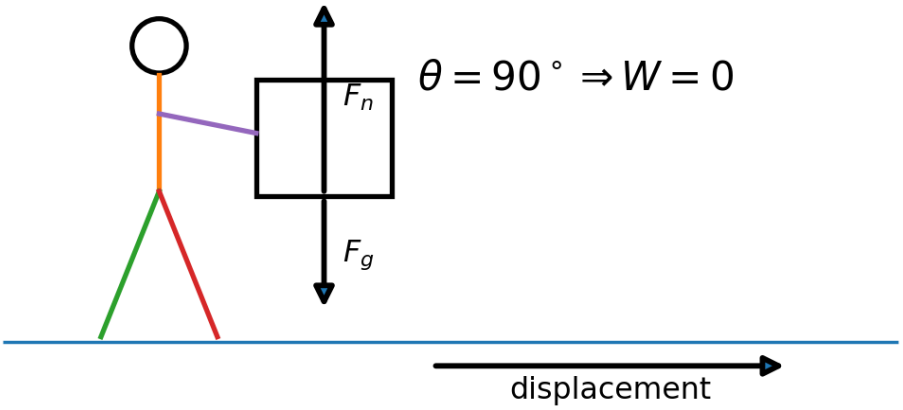
The person then increases the applied force to 540 N. The box again begins at 1.0 m s^{-1}; assume the kinetic-friction force remains unchanged.

(d) Determine the work done by the person over 3.5 m.

(e) Determine the work done by friction on the box.

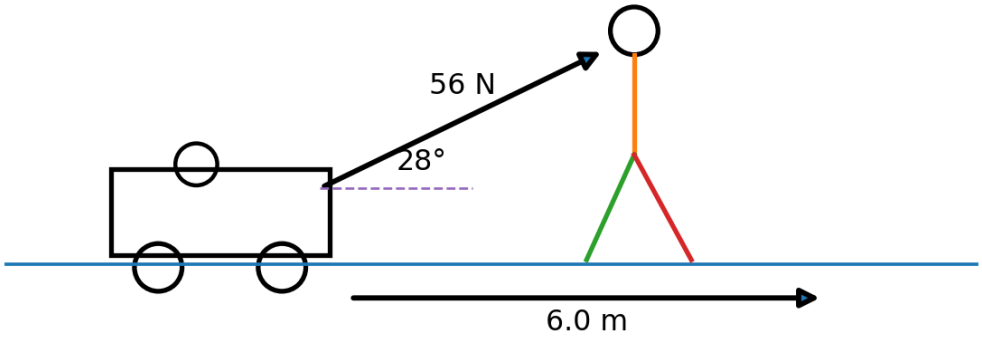
(f) Hence determine the increase in kinetic energy and the final speed.

FRQ 3 · Carrying a box: zero work



A person carries a 25 kg box horizontally at a constant speed of 1.4 m s^{-1}. Determine the mechanical work done on the box by the person's support force as the box moves 16 m horizontally. State any assumptions.

FRQ 4 · Pulling a child and trolley

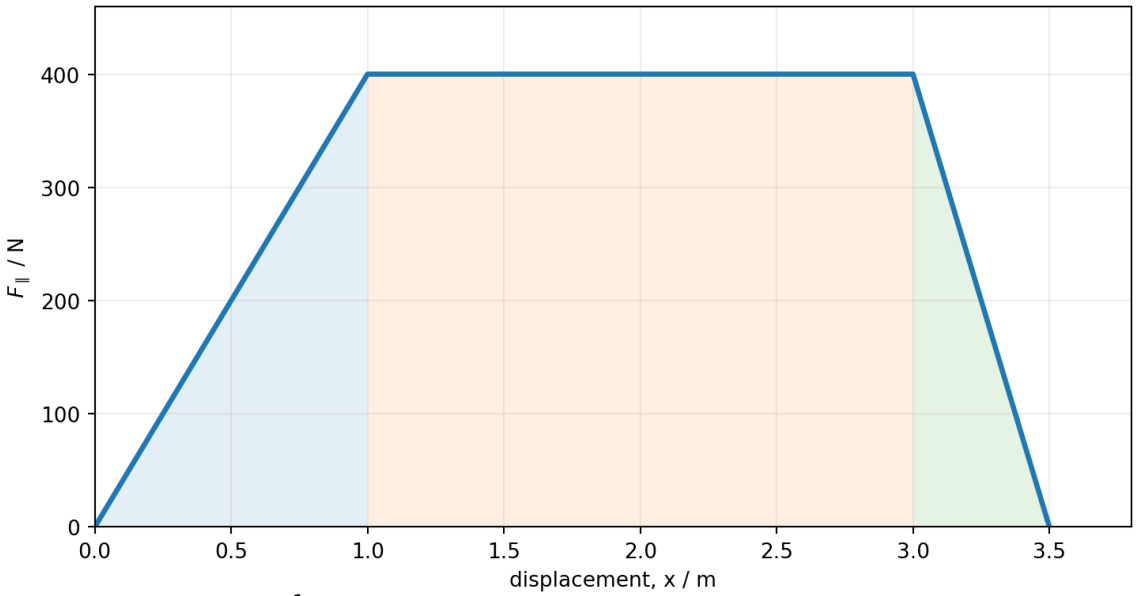


The combined mass of a child and trolley is 42 kg. The father pulls with a constant force of 56 N at 28° above the horizontal. The trolley moves 6.0 m horizontally.

(a) Determine the work done by the father.

(b) If the trolley starts from rest, determine its speed after 6.0 m. State assumptions.

FRQ 5 · Variable net force



A 360 kg box starts at 1.0 m s^{-1}. The graph shows the net force on the box as it moves forward 3.5 m.

(a) Determine the net work done on the box.

(b) Hence determine its speed after 3.5 m.

(c) Discuss what can and cannot be concluded about the work done by the person, and state whether the mechanical energy of the box-Earth system is conserved.

FRQ 6 · Power while pulling at an angle

A student pulls a 20 kg crate across a horizontal floor with a constant force of 90 N at 35° above horizontal. The crate moves at a constant speed of 1.8 m s^{-1} for 12 s.

(a) Determine the work done by the student over the 12 s interval.

(b) Determine the average power delivered by the student.

(c) Determine the instantaneous power delivered by the student and explain why it is constant.

(d) Determine the magnitude of the kinetic-friction force.

FRQ 7 · Experimental / graphical work and power

A cart moves along a horizontal track. A force sensor measures the component of applied force parallel to the cart's displacement, and a motion sensor measures the cart's velocity.

(a) Describe how the student could determine the work done by the applied force from a graph.

(b) Describe how the student could determine the cart's change in kinetic energy and test the work-energy theorem.

(c) At one instant the force sensor reads 3.6 N and the cart speed is 1.25 m s^{-1}. Determine the instantaneous power delivered by the applied force.

(d) The applied force is not the only horizontal force. Explain why the work done by the applied force alone need not equal the change in kinetic energy.